

13.7 Adding and Subtracting Linear Expressions With Integer Coefficients

Lesson Introduction

 Benchmark	MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.		 Learning Target	I can use algebra tiles to add and subtract linear expressions.
 Benchmark Coherence	Building On		Working Toward	
	MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.		MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.	
 Essential Questions	- How can you represent algebraic expressions using algebra tiles? - How do you determine whether terms are like or unlike? - How could you use zero pairs when simplifying expressions?			
 Lesson Narrative	Students simplify algebraic expressions using algebra tiles by combining like terms and eliminating zero pairs. Students have prior experience using integer chips to add and subtract rational numbers. They will connect this knowledge to algebra tiles and combining like terms to write equivalent algebraic expressions.			
 Component Summary	13.7.1 Warm-Up (~5 min.)	Students learn about engineers as a Career Profile (video) (<i>SE p. 295</i>)	13.7.5 Guided Instruction (10 min.)	Model algebraic expressions using algebra tiles, identify like terms, eliminate zero pairs, and write in simplest form (<i>SE p. 299</i>)
	13.7.2 Refresher (10 min.)	Exploring zero pairs using algebra tiles (<i>SE p. 295</i>)	13.7.6 Guided Practice (5 min.)	Practice adding and subtracting expressions with algebra tiles and writing expressions in simplest form (<i>SE p. 300</i>)
	13.7.3 Guided Instruction (5 min.)	Introduction to academic vocabulary related to algebraic expressions (<i>SE p. 296</i>)	13.7.7 Your Turn! (5 min.)	Apply understanding of combining like terms to write equivalent algebraic expressions (<i>SE p. 300</i>)
	13.7.4 Exploration (5-10 min.)	Applying understanding of zero pairs and like terms with algebra tiles to algebraic expressions (<i>SE p. 297</i>)	13.7.8 Wrap-Up (~5 min.)	Demonstrate understanding of the key vocabulary from the lesson (<i>SE p. 301</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Zero pair - Term - Expression - Coefficient - Like terms - Constant <u>Additional Terms:</u> N/A		- Career Profile video - Algebra tiles (concrete and/or virtual) (Optional) Chart paper (Optional) Marker	
			Additional Preparation	
			13.7.1 Warm-Up contains a video to accompany the question prompts (three-minute video). - If using concrete algebra tiles, determine how many tiles are needed to model all of the examples in the lesson and organize student sets in a container or plastic bag.	

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may try to combine terms that are not like terms, for example, $2 + 3x = 5x$ or $3x + 2y = 5xy$. - Students may not recognize the coefficient of 1 when adding like terms, for example, $x + 2x + 3x = 5x$ rather than $6x$.		- Determine if your students need the support of all or part of the refresher. It is possible this component does not need to be used as students did this in Unit 11. - Consider making student anchor charts, Frayer Models, or a word wall to reinforce the vocabulary terms presented in the lesson. - Notice the intentional spiraling throughout these lessons. Students do not need mastery yet, but instead have opportunities to continue practicing adding and subtracting over the next two lessons. - If time is an issue, consider assigning 13.7.7 Your Turn! for homework.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Notice and Wonder Make use of this routine in 13.7.2 Refresher in order to position students to make sense of the rationale behind why variables like x and y are not considered like terms. The visual representation of algebra tiles enhances this routine.		MA.K12.MTR.2.1: Multiple Representations Multiple portions of this lesson provide opportunities for students to represent algebraic expressions in various ways. In 13.7.2 Refresher, students have the opportunity to see multiple ways to represent numeric expressions both using algebra tiles and creating other equivalent expressions.	
 Differentiate Instruction	Consider providing algebra tiles, having students sketch algebra tiles, or providing access to the digital version of these manipulatives as an alternative method to capturing how students are processing the meaning of zero pairs and simplifying expressions.			
Lesson Notes:				

13.7.1 Warm-Up: Career in Robotics

Component Overview: Play the Career Profile video for the entire class. After students watch this two-minute video, prompt the students to read and answer the questions in the Warm-Up independently.



13.7.1 Warm-Up: Career in Robotics

Tessa Lau is the Chief Robot Whisperer for a robotics company. The company creates robot helpers for the service industry. Some of their robots perform tasks like delivering items to guests at a hotel. Tessa has studied applied



physics, computer science, artificial intelligence, and human-computer interaction. She loves hands-on experiments, building things, and the speed of creating with computers. Through programming and engineering, one of her primary jobs is to look at a robot in context and make sure the robots behave well in human environments.

1. Have you ever considered a career as an engineer?
Answers will vary.
2. How do you think a robot could be helpful to you in your everyday life?
Answers will vary.
3. Tessa's advice is "Don't listen to someone who tells you that you can't do something." Describe a time someone told you that you were not able to do something.
Answers will vary.



Do you know someone who is an engineer or computer programmer? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later for project-based learning around this Career Profile.



Ask students to explain why they have or have not considered the profession. Consider specifying how many examples students should come up with. Is one example okay or should students write more than one example. Is a text-based response sufficient or would you like students to include a diagram (of their robot invention and/or the task it performs)?

13.7.2 Refresher: Adding and Subtracting Integers with Algebra Tiles

Component Overview: Students connect prior knowledge of integer chips to using algebra tiles to represent zero pairs and algebraic expressions.



13.7.2 Refresher: Adding and Subtracting Integers with Algebra Tiles

Algebra tiles, like integer chips, can be used to model adding and subtracting integers. Each small square represents a positive 1 (blue) or a negative 1 (red).

Integer Chips	Algebra Tiles
 	 

Adding a positive 1 (blue) and a negative 1 (red) represents a zero pair because $1 + -1 = 0$, as shown.

$$\begin{array}{|c|c|} \hline 1 & -1 \\ \hline \end{array} = 0$$

- Describe why this diagram represents the value 0.

1	-1
1	-1
1	-1

Zero pairs result from the sum of each of the three sets of opposites. Algebraically:

$$1 + (-1) + 1 + (-1) + 1 + (-1) = 0 + 0 + 0 = 0.$$

A **zero pair** is a pair of numbers whose sum is zero.



Choose a student to read the definition of zero pairs out loud and briefly discuss how zero pairs are represented using algebra tiles. Students then complete a table to model numeric expressions with algebra tiles. Before answering question 2b and 2c, consider displaying student work that used zero pairs and student work that did not to elicit a discussion about when utilizing zero pairs is necessary and effective.

- How are the representations of zero pairs with integer chips and algebra tiles the same?
- How are they different?



Consider the different entry points and ways to engage with key vocabulary, such as zero pair. Word walls, vocabulary cards, and sentence stems are all strategies that can serve multiple learning styles with vocabulary content.

13.7.2 Refresher: Adding and Subtracting Integers with Algebra Tiles (continued)

2. Complete the following problems.

a. Complete the table.

Numeric Expression	Algebra Tiles Representation	Value
$-7 + 2$		-5
$*6 - 8$		-2
$*-1 - 5$		-6
$*4 - (-3)$		7

- b. For some of the representations with algebra tiles, zero pairs had to be added to the representation. Put an asterisk (*) next to the expressions where this was the case.
- c. Explain why zero pairs had to be added to the representation in these cases.

For $6 - 8$, two zero pairs were added to be able to subtract positive 8.

For $-1 - 5$, five zero pairs were added to be able to subtract positive 5.

For $4 - (-3)$, three zero pairs were added to be able to subtract negative 3.



Students can either place their algebra tiles in each blank table or sketch a depiction of their algebra tiles that represent each expression.

2. Complete the following problems.

a. Complete the table.

Numeric Expression	Algebra Tiles Representation	Value
$-7 + 2$		



Notice and Wonder

The purpose of this activity is to refresh prior learning with adding and subtracting integers before moving into adding and subtracting algebraic expressions.

- For which expressions did zero pairs need to be added to the model to find their value?
- Why was it necessary to add zero pairs in these cases?



13.7.3 Guided Instruction: Algebra Vocabulary

Component Overview: Students review key definitions necessary for working with algebraic expressions. Read each definition out loud and provide an example, then have students share another example or rephrase the definition in their own words.



13.7.3 Guided Instruction: Algebra Vocabulary

There are some fundamental definitions that are necessary to know for working with algebraic expressions.

A **term** is a single number or variable, or numbers and variables multiplied together. For example, the expression $0.5 - 4y$ has two terms.

The expression $0.5 - 4y$ has two terms.

0.5	-4y
term 1	term 2

One of the terms in this expression is a constant term.

1. Which term is the constant term? Explain.

The term 0.5 is the constant because it is not multiplied by a variable.



An **expression** is a mathematical statement containing numerals, operators, grouping symbols, and symbols or variables for unknown values. An expression does not contain an equal sign or inequality symbol.

For example, $4 + 5$, $3x - 2$, and $6x - 7y + 11$ are all expressions.



A **coefficient** is the number or constant that multiplies a variable in an algebraic expression. If no number is specified, the coefficient is 1.

In the expression $4xy$, 4 is the coefficient.

In the expression x , 1 is the coefficient.

In the expression $-8y$, -8 is the coefficient.

Variable terms have coefficients while constant terms do not.



Consider a variety of strategies to incorporate this math vocabulary into your classroom.

- **Visual Scaffolding:** Learning tools like foldables or a Frayer Model provide examples and nonexamples for students to make sense in context.
- **Positive Affirmation:** Consider using a token economy or form of positive affirmation (snaps) when a student correctly utilizes glossary terms in context when responding.
- **Vocab Captain:** Nominate a different student per day to monitor opportunities in which students could have replaced an informal word with a math glossary term. Consider giving the class a homework pass for each class period in which there were zero misses.
- **Visual Word Wall:** Create a visual word wall where students see the vocabulary term and a corresponding example.

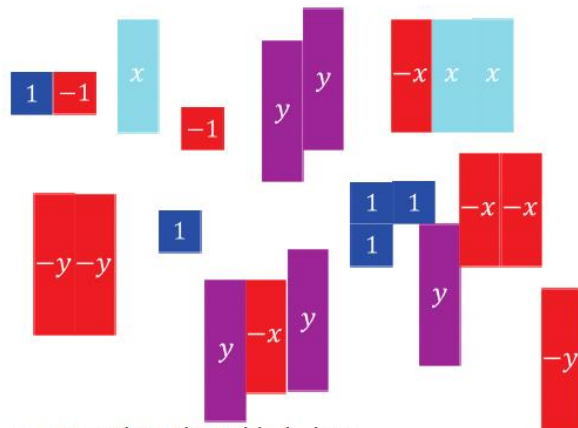
13.7.4 Exploration: What Are Like Terms?

Component Overview: Students categorize algebra tiles and use definitions from 13.7.3 Guided Instruction to create an algebraic expression to match the tiles. Students combine zero pairs to simplify the expression. Listen for students utilizing the key vocabulary introduced in the previous activity.



13.7.4 Exploration: What are Like Terms?

1. Here is a collection of algebra tiles.



- a. Complete the table below.

Type of Algebra Tile	Number of Tiles
x	3
$(-x)$	4
y	5
$(-y)$	3
1	5
(-1)	2

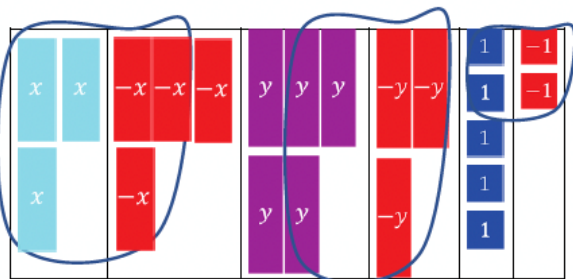


Consider providing algebra tiles for students or groups of students as kinesthetic scaffolding.

13.7.4 Exploration: What Are Like Terms? (continued)

- b. Use the definitions of coefficient and term to complete the expression by writing the coefficient or constant term that matches the number of tiles of each type.

$$3x + (-4x) + 5y + (-3y) + 5 + (-2)$$



- c. Draw the number of algebra tiles of each type under the term in the expression.
- d. Circle the zero pairs in your drawing.
- e. Complete the expression below after eliminating the zero pairs.

$$-x + 2y + 3$$

Like terms are terms with the exact same variable(s) raised to the same exponent.

A **constant** is a fixed numerical value with no variables. All constants are like terms.



Provide space for students to explore and make mistakes as they begin developing their understanding of variables, coefficients, and like terms.

As students explore, monitor for students who might interpret the three light blue algebra tiles as $x + 3$ instead of $3x$. Consider asking those students to describe what $x + 3$ might look like with tiles and compare that to what they see here. (Notice the answer key in 13.7.6 Guided Practice is a sample of what this could look like.)

13.7.4 Exploration: What Are Like Terms? (continued)

2. Answer the following questions about like terms.

a. Classify the following terms.

	Like Terms	Unlike Terms
y and 6	☐	☒
$3x$ and 3	☐	☒
$5x$ and $-3x$	☒	☐
$10, -2, 7.5$	☒	☐



Consider a Turn and Talk for question 2 in which students first answer parts a-c on their own before sharing with a partner to refine their responses.

b. How did you determine whether the terms were like or unlike?

Answers will vary. If the terms had the exact same variable(s) raised to the same exponent, they are like terms. If the terms did not, they were unlike.

c. Compare your answers with your partner.

13.7.5 Guided Instruction: Adding and Subtracting Expressions with Algebra Tiles

Component Overview: Teacher models how to use algebra tiles to eliminate zero pairs and create an equivalent algebraic expression. Students represent a given algebraic expression using algebra tiles and then create an equivalent expression after eliminating zero pairs.



13.7.5 Guided Instruction: Adding and Subtracting Expressions with Algebra Tiles

The algebra tiles shown represent x and $-x$ as rectangles, and integers as squares. Red represents the negative of that variable or constant, while the blue shades represent the positive.



Variables also form zero pairs, as shown.



- Complete the table by crossing out the zero pairs and recording how many there are. Write the resulting algebraic expression after eliminating the zero pairs.

Algebra Tiles	Number of Zero Pairs	Algebraic Expression
Representation: 	2 zero pairs of x	$x - 2$



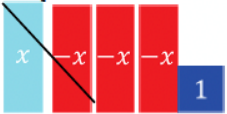
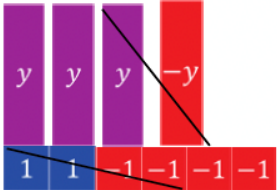
For students new to algebra tiles, consider discussing the dimensions of the tiles, noting that “like” terms have “like” sizes. This would be important to note for any students sketching tiles.



Consider providing algebra tiles for students or groups of students as kinesthetic scaffolding.

13.7.5 Guided Instruction: Adding and Subtracting Expressions with Algebra Tiles (continued)

2. Draw a representation of the following expressions with algebra tiles. Cross out any zero pairs. Then write an equivalent algebraic expression after eliminating any zero pairs.

Algebraic Expression and Representation	Equivalent Algebraic Expression
$x - 3x + 1$ Representation: 	$-2x + 1$
$-4 + 3y - y + 2$ Representation: 	$2y + (-2)$ <i>or</i> $2y - 2$



As students transition from numeric expressions to algebraic expressions, it becomes more essential to highlight the differences in size as an indicator for the like terms. Transition student language from “same size” to “like terms” for effective transitions to algebraic expressions.

13.7.6 Guided Practice: Adding and Subtracting Expressions with Algebra Tiles

Component Overview: Students model given expressions using algebra tiles and determine if they are equivalent.



13.7.6 Guided Practice: Adding and Subtracting Expressions with Algebra Tiles

1. Consider the expressions $2x$ and $x + 2$.
 - a. Draw a model of $2x$ and $x + 2$ with algebra tiles.

Expression	Algebra Tiles Model
$2x$	
$x + 2$	

- b. How are they different?

The expression $2x$ represents the sum of two like terms: $x + x$. Equivalently, it represents the product: $2x$. On the other hand, $x + 2$ represents the sum of two terms that are not like terms, x and 2 , so they cannot be combined.

- c. Are $2x$ and $x + 2$ equivalent algebraic expressions?
No, see part b for justification.

$$x + x \neq x + 2$$



Consider a Turn and Talk or Think-Pair-Share for students or classrooms who learn best with a partner.

- How are these expressions the same? How are they different?
- If the two expressions were added together, what would be the resulting expression?



Consider challenging students to provide mathematical justification by substituting a value for x to prove they are not equivalent.

13.7.7 Your Turn!

Component Overview: Students combine like terms to create equivalent algebraic expressions.



13.7.7 Your Turn!

Write an equivalent expression by combining like terms.

1. $2x + 4x =$ 6x

2. $-y + 7 + 4y - 1 =$ 3y + 6

3. $x - x + x - x + 1 =$ 1

4. $8x - 4 - 4x =$ 4x - 4

5. $2x - 4x =$ -2x



Consider giving students the opportunity to use or sketch algebra tiles or use the digital manipulatives as a method of producing their answers.

13.7.8 Wrap-Up: Cool Down

Component Overview: Students reflect on their progress and provide examples as evidence.



13.7.8 Wrap-Up: Cool Down

- For each key word, check the box in the column that describes your knowledge of the word. Then answer the questions or complete the statements in the last column.

Answers will vary.

Key Word	I already knew the meaning of this word.	I've heard this word before, but I was not sure what it meant.	Questions
Expression	☐	☐	An example of a numeric expression is <u>3</u> . An example of an algebraic expression is <u>$2x + 4$</u> .
Term	☐	☐	In the expression $8x - 9 + 2y$, how many terms are there? <u>3</u>
Coefficient	☐	☐	In the expression $8x - 9 + 2y$, what are the coefficients? <u>8 and 2</u>
Equivalent	☐	☐	When working with algebraic expressions the word "equivalent" means <u>two expressions have the same value when the variables are set to the same numbers</u> .



Consider utilizing students' responses to inform how to scaffold instruction for the next lesson when adding linear expressions.



If students state that -9 is a coefficient, further ask them why. Theoretically it is a coefficient of x^0 . If a student does not use that reasoning, then the student has the misconception that any numeric value is a coefficient.